
Practical week-4

Step 1 — Create Database

The screenshot shows the SQL Developer interface. The Navigator pane on the left displays the schema hierarchy: kldb, p_week1, p_week2, sweek1, and sys. The main query editor contains two lines of SQL: `create database pweek4;` and `use pweek4;`. The Output pane at the bottom shows the execution results:

#	Time	Action	Message	Duration / Fetch
1	14:31:10	create database pweek4	1 row(s) affected	0.016 sec
2	14:31:54	use pweek4	0 row(s) affected	0.000 sec

Step 2 — Create Books Table (DDL + Constraints)

The screenshot shows the SQL Developer interface with the same schema hierarchy. The main query editor contains the following SQL code:

```
1 create database pweek4;
2 use pweek4;
3 CREATE TABLE Books
4 (
5     book_id INT AUTO_INCREMENT PRIMARY KEY,
6
7     title VARCHAR(200) NOT NULL,
8
9     author VARCHAR(150) NOT NULL,
10
11     genre VARCHAR(100),
12
13     description TEXT,
14
15     published_year INT
16 );
17
```

The Output pane at the bottom shows the execution results:

#	Time	Action	Message	Duration / Fetch
1	14:31:10	create database pweek4	1 row(s) affected	0.016 sec
2	14:31:54	use pweek4	0 row(s) affected	0.000 sec
3	14:33:54	CREATE TABLE Books (book_id INT AUTO_INCREME...	0 row(s) affected	0.031 sec

Step 3 — Describe Books Table

The screenshot shows a SQL IDE interface with a query editor, a schema navigator, and an output window. The query editor contains the following SQL code:

```
3 CREATE TABLE Books
4 (
5     book_id INT AUTO_INCREMENT PRIMARY KEY,
6
7     title VARCHAR(200) NOT NULL,
8
9     author VARCHAR(150) NOT NULL,
10 )
```

The schema navigator on the left shows a database named 'kldb' with several schemas, including 'p_week1'. The output window at the bottom shows the execution results of the query:

#	Time	Action	Message	Duration / Fetch
1	14:31:10	create database pweek4	1 row(s) affected	0.016 sec
2	14:31:54	use pweek4	0 row(s) affected	0.000 sec
3	14:33:54	CREATE TABLE Books (book_id INT AUTO_INCREM...	0 row(s) affected	0.031 sec
4	14:36:29	describe Books	6 row(s) returned	0.000 sec / 0.000 sec

The 'describe Books' output is also displayed in a table format in the center of the IDE:

Field	Type	Null	Key	Default	Extra
book_id	int	NO	PRI	<u>NONE</u>	auto_increment
title	varchar(200)	NO		<u>NONE</u>	
author	varchar(150)	NO		<u>NONE</u>	
genre	varchar(100)	YES		<u>NONE</u>	
description	text	YES		<u>NONE</u>	
published_year	int	YES		<u>NONE</u>	

Step 4 — Insert 10 Books and Display

The screenshot shows the same SQL IDE interface, but now with an 'INSERT INTO Books' query in the query editor. The query is as follows:

```
16 );
17 describe Books;
18 INSERT INTO Books
19 (
20     title,
21     author,
22     genre,
23     description,
24     published_year
25 )
26 VALUES
27 (
28     'The Martian',
29     'Andy Weir',
30     'Science Fiction',
31     'An astronaut struggles to survive alone on Mars.',
32     2011
33 ),
34 )
```

The output window at the bottom shows the execution results of the query:

#	Time	Action	Message	Duration / Fetch
1	14:31:10	create database pweek4	1 row(s) affected	0.016 sec
2	14:31:54	use pweek4	0 row(s) affected	0.000 sec
3	14:33:54	CREATE TABLE Books (book_id INT AUTO_INCREM...	0 row(s) affected	0.031 sec
4	14:36:29	describe Books	6 row(s) returned	0.000 sec / 0.000 sec
5	14:39:56	INSERT INTO Books (title, author, genre, descr...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.016 sec

Step 5 — Add Embedding Column (JSON)

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

```
97 • describe Books;
98 • SELECT
99     book_id,
100     title,
101     author,
102     genre,
103     published_year
104 FROM Books
105 ORDER BY book_id;
```

book_id	title	author	genre
1	The Martian	Andy Weir	Science Fiction
2	Dune	Frank Herbert	Science Fiction
3	The Hitchhikers Guide to the Galaxy	Douglas Adams	Science Fiction
4	Pride and Prejudice	Jane Austen	Romance
5	The Great Gatsby	F. Scott Fitzgerald	Classic
6	Foundation	Isaac Asimov	Science Fiction
7	The Hobbit	J.R.R. Tolkien	Fantasy
8	Enders Game	Orson Scott Card	Science Fiction

Schema: p_week1

Output: Action Output

#	Time	Action	Message	Duration / Fetch
2	14:31:54	use pweek4	0 row(s) affected	0.000 sec
3	14:33:54	CREATE TABLE Books (book_id INT AUTO_INCRE...	0 row(s) affected	0.031 sec
4	14:36:29	describe Books	6 row(s) returned	0.000 sec / 0.000 sec
5	14:39:56	INSERT INTO Books (title, author, genre, des...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.016 sec
6	14:40:58	describe Books	6 row(s) returned	0.000 sec / 0.000 sec
7	14:42:00	SELECT book_id, title, author, genre, publis...	10 row(s) returned	0.000 sec / 0.015 sec

Step 6 — Update Book Embeddings

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

```
91     'Twenty Thousand Leagues Under the Seas',
92     'Jules Verne',
93     'Adventure',
94     'A fantastic underwater adventure aboard a mysterious submarine.',
95     1870
96 );
97 • describe Books;
98 • SELECT
99     book_id,
100     title,
101     author,
102     genre,
103     published_year
104 FROM Books
105 ORDER BY book_id;
106
107 • ALTER TABLE Books
108 ADD COLUMN embedding JSON;
```

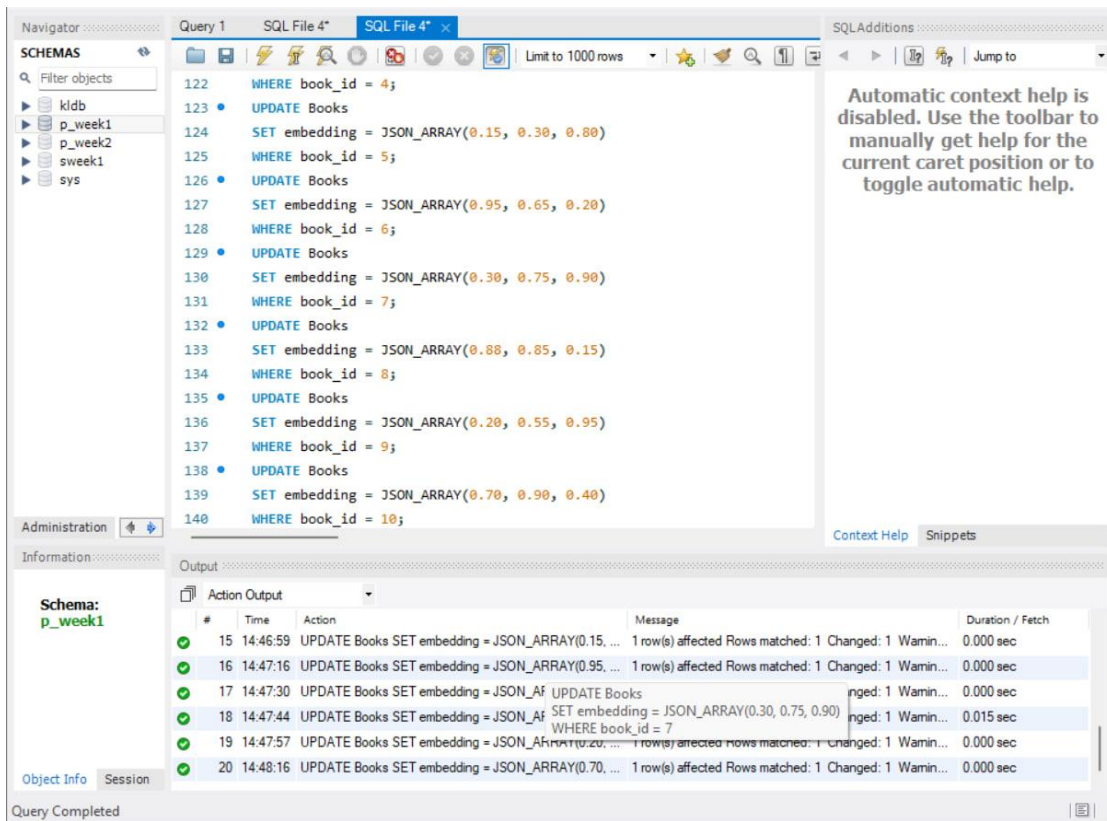
Schema: p_week1

Output: Action Output

#	Time	Action	Message	Duration / Fetch
3	14:33:54	CREATE TABLE Books (book_id INT AUTO_INCRE...	0 row(s) affected	0.031 sec
4	14:36:29	describe Books	6 row(s) returned	0.000 sec / 0.000 sec
5	14:39:56	INSERT INTO Books (title, author, genre, des...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.016 sec
6	14:40:58	describe Books	6 row(s) returned	0.000 sec / 0.000 sec
7	14:42:00	SELECT book_id, title, author, genre, publis...	10 row(s) returned	0.000 sec / 0.015 sec
8	14:43:00	ALTER TABLE Books ADD COLUMN embedding JSON	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.031 sec

Query Completed

Step 7 — Display Book Embeddings



Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

```
122 WHERE book_id = 4;
123 • UPDATE Books
124 SET embedding = JSON_ARRAY(0.15, 0.30, 0.80)
125 WHERE book_id = 5;
126 • UPDATE Books
127 SET embedding = JSON_ARRAY(0.95, 0.65, 0.20)
128 WHERE book_id = 6;
129 • UPDATE Books
130 SET embedding = JSON_ARRAY(0.30, 0.75, 0.90)
131 WHERE book_id = 7;
132 • UPDATE Books
133 SET embedding = JSON_ARRAY(0.88, 0.85, 0.15)
134 WHERE book_id = 8;
135 • UPDATE Books
136 SET embedding = JSON_ARRAY(0.20, 0.55, 0.95)
137 WHERE book_id = 9;
138 • UPDATE Books
139 SET embedding = JSON_ARRAY(0.70, 0.90, 0.40)
140 WHERE book_id = 10;
```

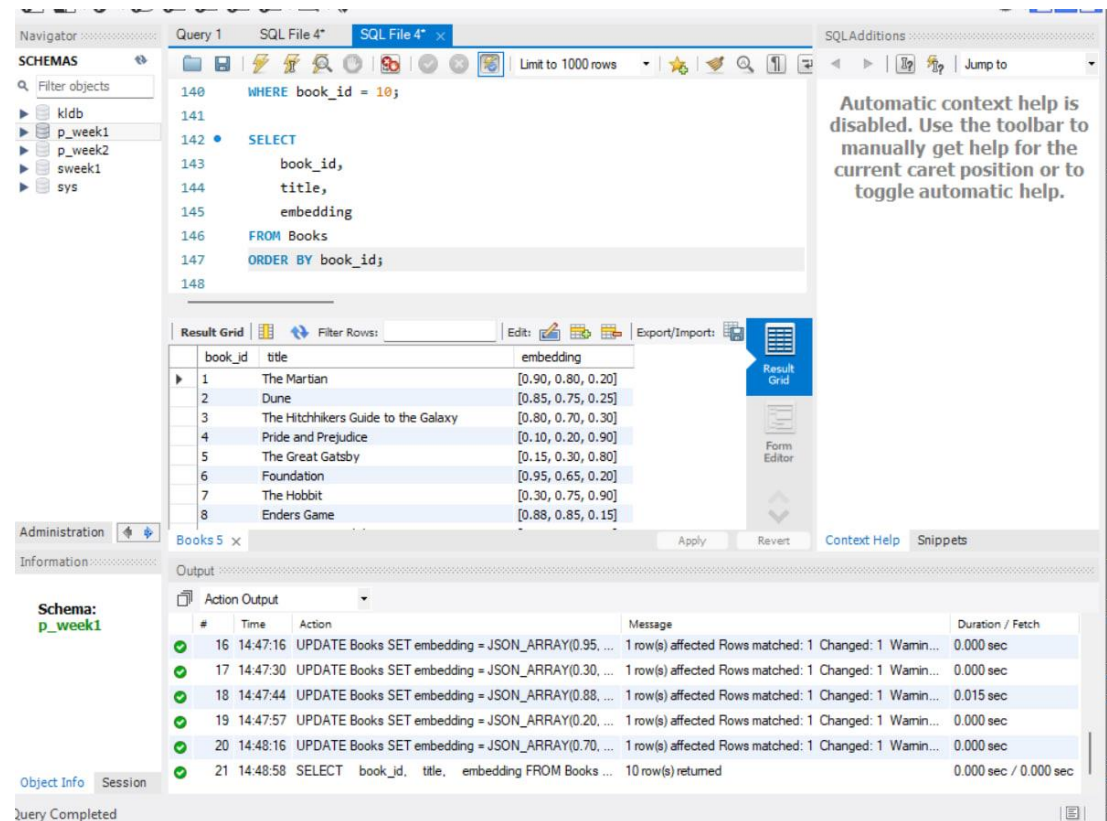
Schema: p_week1

Output

#	Time	Action	Message	Duration / Fetch
✓ 15	14:46:59	UPDATE Books SET embedding = JSON_ARRAY(0.15, ...	1 row(s) affected Rows matched: 1 Changed: 1 Wamin...	0.000 sec
✓ 16	14:47:16	UPDATE Books SET embedding = JSON_ARRAY(0.95, ...	1 row(s) affected Rows matched: 1 Changed: 1 Wamin...	0.000 sec
✓ 17	14:47:30	UPDATE Books SET embedding = JSON_ARRAY(0.30, ...	1 row(s) affected Rows matched: 1 Changed: 1 Wamin...	0.000 sec
✓ 18	14:47:44	UPDATE Books SET embedding = JSON_ARRAY(0.88, ...	1 row(s) affected Rows matched: 1 Changed: 1 Wamin...	0.015 sec
✓ 19	14:47:57	UPDATE Books SET embedding = JSON_ARRAY(0.20, ...	1 row(s) affected Rows matched: 1 Changed: 1 Wamin...	0.000 sec
✓ 20	14:48:16	UPDATE Books SET embedding = JSON_ARRAY(0.70, ...	1 row(s) affected Rows matched: 1 Changed: 1 Wamin...	0.000 sec

Query Completed

Step 8 — L2 Distance Query



Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

```
140 WHERE book_id = 10;
141
142 • SELECT
143 book_id,
144 title,
145 embedding
146 FROM Books
147 ORDER BY book_id;
148
```

Schema: p_week1

Result Grid

book_id	title	embedding
1	The Martian	[0.90, 0.80, 0.20]
2	Dune	[0.85, 0.75, 0.25]
3	The Hitchhikers Guide to the Galaxy	[0.80, 0.70, 0.30]
4	Pride and Prejudice	[0.10, 0.20, 0.90]
5	The Great Gatsby	[0.15, 0.30, 0.80]
6	Foundation	[0.95, 0.65, 0.20]
7	The Hobbit	[0.30, 0.75, 0.90]
8	Enders Game	[0.88, 0.85, 0.15]

Output

#	Time	Action	Message	Duration / Fetch
✓ 16	14:47:16	UPDATE Books SET embedding = JSON_ARRAY(0.95, ...	1 row(s) affected Rows matched: 1 Changed: 1 Wamin...	0.000 sec
✓ 17	14:47:30	UPDATE Books SET embedding = JSON_ARRAY(0.30, ...	1 row(s) affected Rows matched: 1 Changed: 1 Wamin...	0.000 sec
✓ 18	14:47:44	UPDATE Books SET embedding = JSON_ARRAY(0.88, ...	1 row(s) affected Rows matched: 1 Changed: 1 Wamin...	0.015 sec
✓ 19	14:47:57	UPDATE Books SET embedding = JSON_ARRAY(0.20, ...	1 row(s) affected Rows matched: 1 Changed: 1 Wamin...	0.000 sec
✓ 20	14:48:16	UPDATE Books SET embedding = JSON_ARRAY(0.70, ...	1 row(s) affected Rows matched: 1 Changed: 1 Wamin...	0.000 sec
✓ 21	14:48:58	SELECT book_id, title, embedding FROM Books ...	10 row(s) returned	0.000 sec / 0.000 sec

Query Completed

Step 9 — Cosine Similarity Query

The screenshot shows the SQL Developer interface with a query window titled 'Query 1' containing the following SQL code:

```
SELECT
    book_id,
    title,
    author,
    genre,
    ROUND(
        SQRT(
            POW(
```

The 'Result Grid' displays the following data:

book_id	title	author	genre	l2_distance
1	The Martian	Andy Weir	Science Fiction	0
8	Enders Game	Orson Scott Card	Science Fiction	0.0735
2	Dune	Frank Herbert	Science Fiction	0.0866

The 'Output' pane shows the following log entries:

#	Time	Action	Message
17	14:47:30	UPDATE Books SET embedding = JSON_ARRAY(0.30, ...	1 row(s) affected Rows matched: 1
18	14:47:44	UPDATE Books SET embedding = JSON_ARRAY(0.88, ...	1 row(s) affected Rows matched: 1
19	14:47:57	UPDATE Books SET embedding = JSON_ARRAY(0.20, ...	1 row(s) affected Rows matched: 1
20	14:48:16	UPDATE Books SET embedding = JSON_ARRAY(0.70, ...	1 row(s) affected Rows matched: 1
21	14:48:58	SELECT book_id, title, embedding FROM Books ...	10 row(s) returned
22	14:52:37	SELECT book_id, title, author, genre, ROUN...	3 row(s) returned

Query Completed

Step 10 — Final Similarity Result

The screenshot shows the SQL Developer interface with a query window titled 'Query 1' containing the following SQL code:

```
genre,
ROUND(
(
    CAST(
        JSON_EXTRACT(embedding, '$[0]')
        AS DECIMAL(10,6)
    ) * 0.90
    +
    CAST(
```

The 'Result Grid' displays the following data:

book_id	title	author	genre	cosine_similarity
1	The Martian	Andy Weir	Science Fiction	1
2	Dune	Frank Herbert	Science Fiction	0.9986
8	Enders Game	Orson Scott Card	Science Fiction	0.9983

The 'Output' pane shows the following log entries:

#	Time	Action	Message
18	14:47:44	UPDATE Books SET embedding = JSON_ARRAY(0.88, ...	1 row(s) affected F
19	14:47:57	UPDATE Books SET embedding = JSON_ARRAY(0.20, ...	1 row(s) affected F
20	14:48:16	UPDATE Books SET embedding = JSON_ARRAY(0.70, ...	1 row(s) affected F
21	14:48:58	SELECT book_id, title, embedding FROM Books ...	10 row(s) returned
22	14:52:37	SELECT book_id, title, author, genre, ROUN...	3 row(s) returned
23	14:54:24	SELECT book_id, title, author, genre, RO...	3 row(s) returned

Query Completed